

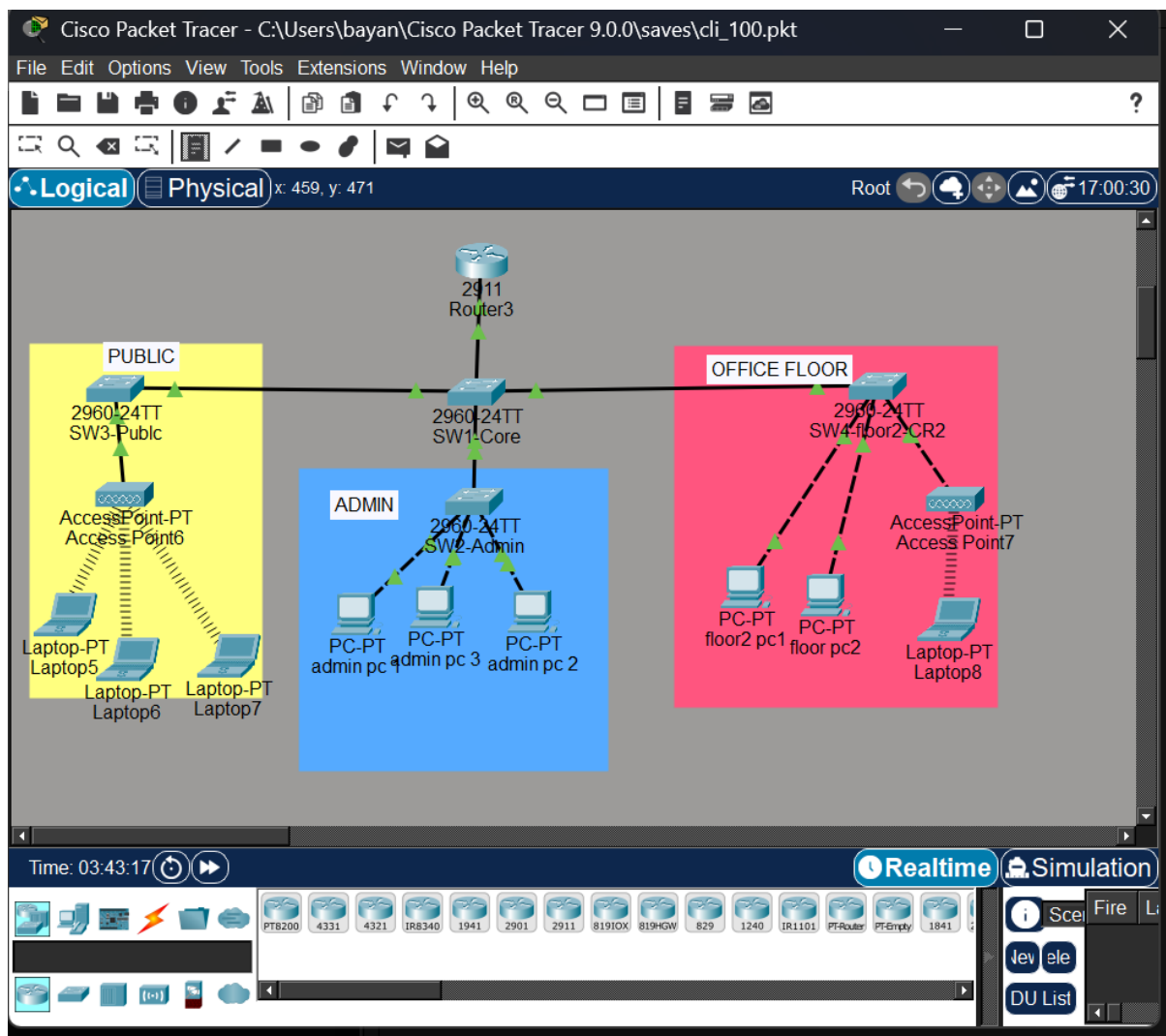
1. Client Requirements

- **Client Profile:** Ratlou Local Municipality Offices (Setlagole).
- **Industry Context:** Municipal Services / Local Government Office.
- **Core Objectives:** Design and deploy a segmented, secure, and scalable local area network for administrative staff, public Wi-Fi access, and future infrastructure expansion using an assigned /24 root addressing block.
- **Design Constraints:**
 - Logical traffic separation between administrative data (VLAN 10), public access (VLAN 20), expansion areas (VLAN 30), and management traffic (VLAN 99).
 - Wireless access strictly restricted to designated public zones and authenticated expansion areas.
 - Router-based dynamic IP address assignment across all operational subnets via DHCP pools.

2. Physical Topology

Hierarchical Topology Model: Core/Distribution layer handled by SW1-Core connected directly to edge access switches (SW2-Admin, SW3-Public, and SW4-Floor2-CR2).

- **Hardware Inventory:**
 - **Router:** R1-Ratlou (Cisco 2911 / ISR series)
 - **Core Switch:** SW1-Core (Cisco Catalyst 2960)
 - **Access Switches:** SW2-Admin, SW3-Public, SW4-Floor2-CR2 (Cisco Catalyst 2960)
 - **Wireless Access Points:** Public AP (AP-PT) and Floor 2 WAP (AP-Floor2)
 - **End-User Devices:** Administrative PCs, Wireless Laptops, and Public PCs/Tablets
- **Physical Cabling & Links:**
 - High-speed FastEthernet and GigabitEthernet trunk links connecting R1-Ratlou to SW1-Core and SW1-Core to edge switches (SW2-Admin, SW3-Public, SW4-Floor2-CR2).
 - Direct copper media connecting end PCs and Access Points to designated switch access ports.



3. Logical Topology

- **Hierarchical Architecture:** Core/Distribution layer handled by SW1-Core connected directly to access layer switches.

VLAN Database Verification: Output from **show vlan brief** on SW1-Core:



```
SW1-Core con0 is now available

Press RETURN to get started.

SW1-Core>enable
SW1-Core#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                           Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24, Gig0/2
10   Admin-Staff             active
20   Public-Area             active
30   CR2-New-Floor           active
99   Management              active
1002 fddi-default            active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
SW1-Core#
```

- **Inter-VLAN Routing:** Router-on-a-stick architecture configured on router R1-Ratlou sub-interfaces using IEEE 802.1Q encapsulation.

DHCP Verification: Output from **Show ip dhcp binding** on R1-Ratlou:



```
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wvl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FX152400RS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

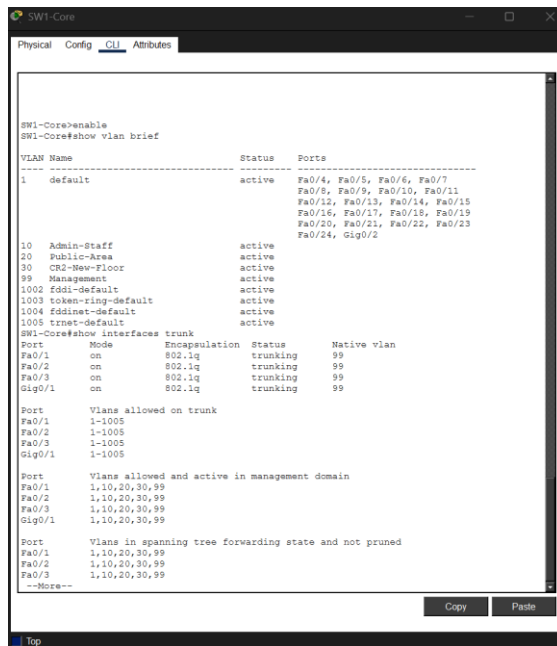
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up

R1-Ratlou>enable
R1-Ratlou#show ip dhcp binding

IP address                Client-ID/          Lease expiration        Type
                           Hardware address
192.168.44.7               00D0.FF91.D3C5      --                        Automatic
192.168.44.6               00E0.F77B.11C0      --                        Automatic
192.168.44.72              0030.A3ED.7513      --                        Automatic
192.168.44.73              0001.43E9.8631      --                        Automatic
192.168.44.71              000C.CF21.07AC      --                        Automatic
192.168.44.75              00E0.8FB1.5294      --                        Automatic
192.168.44.104             0006.2A00.6188      --                        Automatic
192.168.44.103             00E0.8FB1.5294      --                        Automatic
R1-Ratlou#
```

- **Trunking & Native VLAN:** IEEE 802.1Q trunking enabled across core switch links with native VLAN 99 alignment for administrative security.

Trunk Link Verification: Output from **show interfaces trunk:**



```
SW1-Core#show vlan brief
VLAN Name                Status Ports
-----
1    default              active Fa0/4, Fa0/5, Fa0/6, Fa0/7
    Fa0/8, Fa0/9, Fa0/10, Fa0/11
    Fa0/12, Fa0/13, Fa0/14, Fa0/15
    Fa0/16, Fa0/17, Fa0/18, Fa0/19
    Fa0/20, Fa0/21, Fa0/22, Fa0/23
    Fa0/24, Gig0/2
10   Admin-Staff          active
20   Public-Area          active
30   CR2-New-Floor         active
99   Management            active
1002 fddi-default          active
1003 token-ring-default   active
1004 fddinet-default      active
1005 trinet-default       active

SW1-Core#show interfaces trunk
Port      Mode      Encapsulation  Status  Native vlan
Fa0/1     on        802.1q         trunking 99
Fa0/2     on        802.1q         trunking 99
Fa0/3     on        802.1q         trunking 99
Gig0/1    on        802.1q         trunking 99

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/2     1-1005
Fa0/3     1-1005
Gig0/1    1-1005

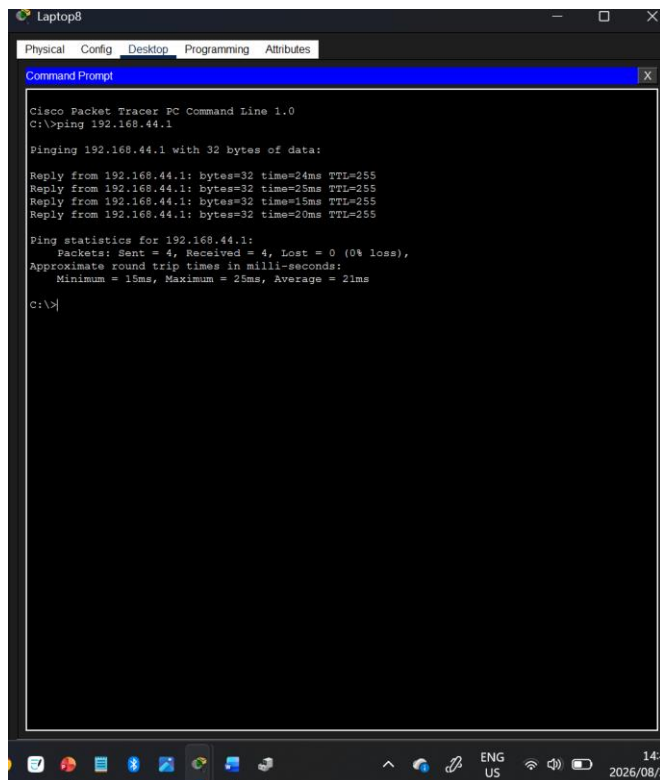
Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,99
Fa0/2     1,10,20,30,99
Fa0/3     1,10,20,30,99
Gig0/1    1,10,20,30,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30,99
Fa0/2     1,10,20,30,99
Fa0/3     1,10,20,30,99
--More--
```

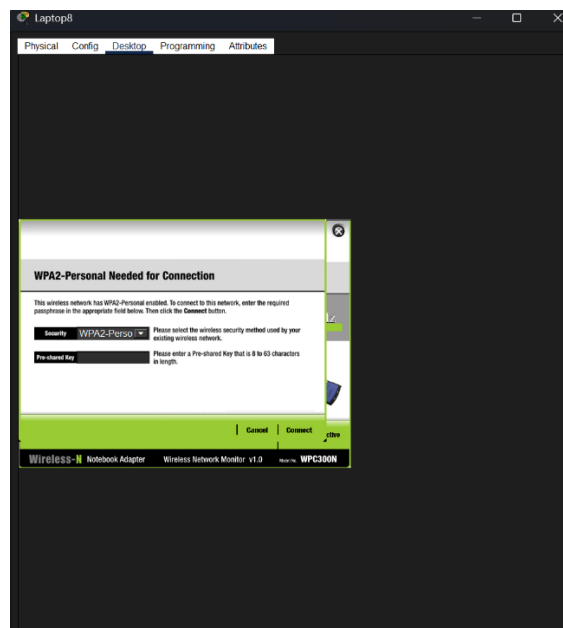
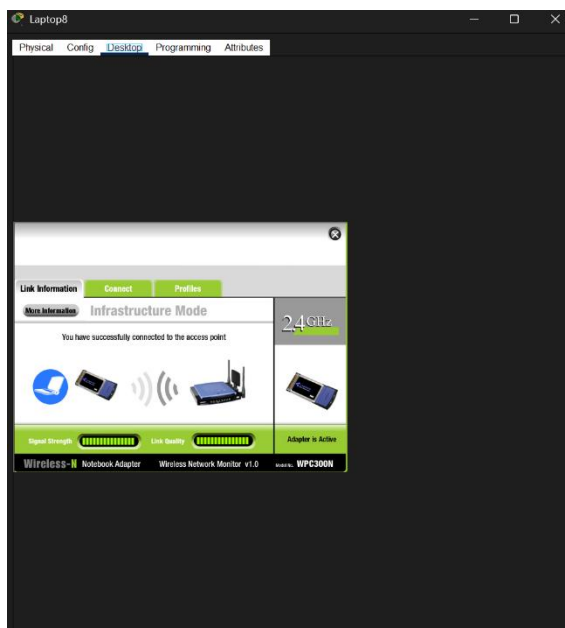
- **Logical Segmentation:**

- **VLAN 10 (Admin-Staff):** Dedicated to internal administrative workstations and database access.
- **VLAN 20 (Public-Area):** Dedicated exclusively to public Wi-Fi access points.
- **VLAN 30 (CR2-New-Floor):** Dedicated to secondary floor wired and authenticated wireless infrastructure.

End-to-End Ping Verification: Successful ping output from Floor 2 Wireless Laptop to Gateway **192.168.44.1**:



WPA2-PSK Authentication for Wireless Office floor laptop:



- **VLAN 99 (Management):** Reserved for native VLAN traffic and network infrastructure management.

4. IP Addressing Plan

Root Network Block: 192.168.44.0/24

Subnet / Function	VLAN ID	Network Address	Subnet Mask	Usable Host Range	Default Gateway
Admin & Staff	VLAN 10	192.168.44.0/26	255.255.255.192	192.168.44.1 - 192.168.44.62	192.168.44.1
Public Wi-Fi Zone	VLAN 20	192.168.44.64/27	255.255.255.224	192.168.44.65 - 192.168.44.94	192.168.44.65
CR2 Expansion Floor	VLAN 30	192.168.44.96/27	255.255.255.224	192.168.44.97 - 192.168.44.126	192.168.44.97
Management Subnet	VLAN 99	192.168.44.128/28	255.255.255.240	192.168.44.129 - 192.168.44.142	192.168.44.129

5. Initial GitHub Repository

- **Repository URL:** https://github.com/Forestgump28/Semester-project_Milestone1.git
- **Repository Assets:**
 - CMPG325_Milestone1_44027516.pkt: Active Cisco Packet Tracer project topology.
 - CMPG325_Milestone1_44027516.pdf: Technical design review report.
 - README.md: Executive summary, VLSM scheme, and progress logs.